



GREAT version 4.0.4 current (08/19/2019 to now)

Warning: Your set hits a large fraction of the genes in the genome, which often does not work well with the GREAT Significant by Both view due to a saturation of the gene-based hypergeometric test. See our [tips for handling large datasets](#) or try the [Significant By Region-based Binomial](#) view.

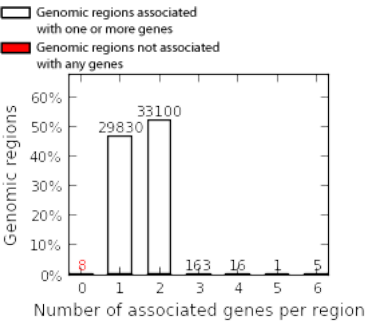
+ Job Description

+ Region-Gene Association Graphs

What do these graphs illustrate?

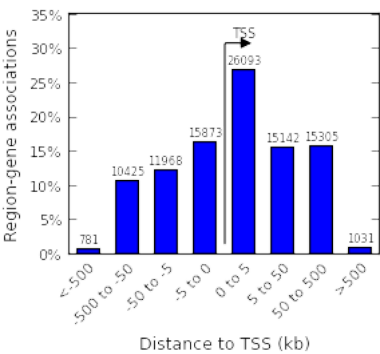
Number of associated genes per region

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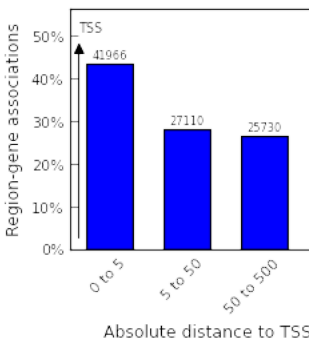
Binned by orientation and distance to TSS

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+ Global Controls

Global Export

Which data is exported by each option?

+ Ensembl Genes (no terms)

- GO Biological Process (20+ terms)

Table controls: Export

Shown top rows in this table: 20 Set

Term annotation count: Min: 1 Max: Inf Set

Visualize this table:

Term Name	Binom Rank	Binom Raw P-Value	Binom FDR Q-Val	Binom Fold Enrichment	Binom Observed Region Hits	Binom Region Set Coverage	Hyper Rank	Hyper FDR Q-Val	Hyper Fold Enrichment	Hyper Observed Gene Hits
nuclear-transcribed mRNA catabolic process	96	3.9716e-67	5.4155e-65	2.0125	755	1.20%	488	2.2070e-10	1.3632	95
retrograde transport, endosome to Golgi	101	1.6266e-64	2.1082e-62	2.3967	490	0.78%	866	6.3995e-6	1.3314	66
cellular response to topologically incorrect protein	113	5.9653e-61	6.9102e-59	2.1814	564	0.89%	597	7.8388e-9	1.3736	75
negative regulation of response to endoplasmic reticulum stress	120	2.4210e-57	2.6410e-55	2.5166	395	0.63%	1,207	4.3784e-4	1.3562	38
mitochondrial transmembrane transport	135	6.3663e-54	6.1729e-52	2.5191	370	0.59%	714	4.3744e-7	1.3919	53
response to cadmium ion	149	6.4460e-52	5.6630e-50	2.2700	438	0.69%	1,509	3.5656e-3	1.2676	51
regulation of mitochondrial membrane permeability	153	2.7658e-51	2.3663e-49	2.2811	428	0.68%	714	4.3744e-7	1.3919	53
endoplasmic reticulum unfolded protein response	163	9.0044e-49	7.2312e-47	2.2149	433	0.69%	888	1.0510e-5	1.3651	51
regulation of membrane permeability	164	9.9724e-49	7.9597e-47	2.1052	486	0.77%	637	4.8093e-8	1.3919	60
chromatin silencing at rDNA	185	1.4313e-44	1.0127e-42	2.9815	228	0.36%	1,284	9.2113e-4	1.3084	47
cellular response to unfolded protein	188	1.7649e-44	1.2289e-42	2.0762	457	0.72%	792	1.7885e-6	1.3679	57
regulation of endoplasmic reticulum stress-induced intrinsic apoptotic signaling pathway	189	1.7812e-44	1.2336e-42	2.6043	285	0.45%	1,329	1.2976e-3	1.3919	27
glutathione metabolic process	191	2.8565e-44	1.9577e-42	2.5035	305	0.48%	1,433	2.4391e-3	1.2848	48
positive regulation of nuclear-transcribed mRNA catabolic process, deadenylation-dependent decay	200	1.6626e-42	1.0882e-40	2.6590	262	0.42%	2,010	2.3512e-2	1.3919	17
positive regulation of mRNA catabolic process	202	3.5623e-42	2.3085e-40	2.4024	314	0.50%	1,856	1.4500e-2	1.3384	25
chaperone-mediated protein folding	209	8.5520e-41	5.3562e-39	2.3149	327	0.52%	722	5.5494e-7	1.3695	61

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Bejerano Lab, Stanford

Name	Rank	P-Value	FDR Q-Val	Enrichment	Region Hits	Coverage	Rank	FDR Q-Val	Enrichment	Gene Hits
intrinsic apoptotic signaling pathway in response to DNA damage by p53 class mediator	211	1.5595e-40	9.6748e-39	2.5488	269	0.43%	1,524	3.9629e-3	1.3470	30
execution phase of apoptosis	218	1.9798e-39	1.1888e-37	2.5246	266	0.42%	1,869	1.5335e-2	1.2820	35
nuclear-transcribed mRNA catabolic process, deadenylation-dependent decay	222	9.6635e-39	5.6980e-37	2.1184	377	0.60%	1,466	2.8859e-3	1.3205	37
cytoplasmic translation	224	1.3712e-38	8.0128e-37	2.4378	278	0.44%	1,575	4.7627e-3	1.2783	45

The test set of 63,123 genomic regions picked 15,371 (72%) of all 21,395 genes.
GO Biological Process has 13,090 terms covering 17,925 (84%) of all 21,395 genes, and 1,163,819 term - gene associations.
13,090 ontology terms (100%) were tested using an annotation count range of [1, Inf].

GO Cellular Component (20+ terms)

Table controls:

Export

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Set

Term annotation count: Min: 1

Max: Inf

Set

Visualize this table: [select one]

Term Name	Binom Rank	Binom Raw P-Value	Binom FDR Q-Val	Binom Fold Enrichment	Binom Observed Region Hits	Binom Region Set Coverage	Hyper Rank	Hyper FDR Q-Val	Hyper Fold Enrichment	Hyper Observed Gene Hits
intrinsic component of mitochondrial membrane	89	3.8301e-38	7.2902e-37	2.0946	381	0.60%	196	1.1584e-5	1.3277	62
integral component of mitochondrial membrane	91	7.4379e-37	1.3846e-35	2.0794	374	0.59%	199	2.0218e-5	1.3256	60
nuclear ubiquitin ligase complex	123	6.4958e-28	8.9463e-27	2.1936	246	0.39%	232	2.1674e-4	1.3571	39
CCR4-NOT complex	130	1.2461e-24	1.6238e-23	2.6483	148	0.23%	355	2.3987e-2	1.3919	16
anaphase-promoting complex	139	1.9596e-23	2.3882e-22	2.7430	132	0.21%	371	3.0344e-2	1.3286	21
MLL1 complex	142	9.4620e-23	1.1288e-21	2.2427	188	0.30%	297	4.7991e-3	1.3439	28
Golgi-associated vesicle membrane	145	3.0065e-22	3.5125e-21	2.0184	235	0.37%	305	5.5073e-3	1.3124	33
DNA-directed RNA polymerase III complex	152	2.8448e-21	3.1705e-20	3.2863	90	0.14%	330	1.3311e-2	1.3919	18
mitochondrial small ribosomal subunit	162	4.1976e-20	4.3893e-19	2.2839	158	0.25%	257	6.2357e-4	1.3919	28
intrinsic component of mitochondrial outer membrane	163	4.9570e-20	5.1517e-19	2.2282	166	0.26%	334	1.4011e-2	1.3362	24
integral component of mitochondrial outer membrane	171	4.5118e-19	4.4696e-18	2.2048	161	0.26%	348	1.8045e-2	1.3339	23
COPI-coated vesicle	173	5.6314e-19	5.5142e-18	2.1933	162	0.26%	299	5.4420e-3	1.3919	21
telomerase holoenzyme complex	188	7.7393e-18	6.9736e-17	2.4209	123	0.19%	299	5.4420e-3	1.3919	21
INO80-type complex	206	1.3941e-16	1.1464e-15	2.4594	110	0.17%	299	5.4420e-3	1.3919	21
TRAPP complex	214	2.2605e-16	1.7894e-15	2.6749	93	0.15%	395	4.1788e-2	1.3919	14
BLOC-1 complex	247	4.5348e-14	3.1101e-13	2.5252	87	0.14%	375	3.1615e-2	1.3919	15
COPI-coated vesicle membrane	276	3.0215e-12	1.8545e-11	2.1369	105	0.17%	375	3.1615e-2	1.3919	15
Ino80 complex	277	3.0725e-12	1.8790e-11	2.5103	75	0.12%	395	4.1788e-2	1.3919	14
DNA helicase complex	279	3.2453e-12	1.9704e-11	2.4561	78	0.12%	375	3.1615e-2	1.3919	15
pre-autophagosomal structure membrane	285	7.8584e-12	4.6709e-11	2.0765	108	0.17%	355	2.3987e-2	1.3919	16

The test set of 63,123 genomic regions picked 15,371 (72%) of all 21,395 genes.
GO Cellular Component has 1,694 terms covering 19,074 (89%) of all 21,395 genes, and 371,380 term - gene associations.
1,694 ontology terms (100%) were tested using an annotation count range of [1, Inf].

GO Molecular Function (12 terms)

Table controls:

Export

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Term annotation count: Min: 1

Max: Inf

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Visualize this table: [select one]

Term Name	Binom Rank	Binom Raw P-Value	Binom FDR Q-Val	Binom Fold Enrichment	Binom Observed Region Hits	Binom Region Set Coverage	Hyper Rank	Hyper FDR Q-Val	Hyper Fold Enrichment	Hyper Observed Gene Hits
rRNA binding	22	1.2054e-78	2.2635e-76	3.3274	349	0.55%	175	5.6065e-5	1.3256	60
unfolded protein binding	25	6.6329e-70	1.0960e-67	2.5344	478	0.76%	236	9.6640e-4	1.2544	73
mitogen-activated protein kinase binding	56	3.1993e-35	2.3601e-33	2.4711	246	0.39%	238	1.1791e-3	1.3919	29
snoRNA binding	76	3.7030e-29	2.0128e-27	2.5635	188	0.30%	238	1.1791e-3	1.3919	29
snRNA binding	84	5.3290e-27	2.6207e-25	2.3437	206	0.33%	350	2.5842e-2	1.2820	35
DNA polymerase binding	86	4.7060e-26	2.2605e-24	3.2869	112	0.18%	356	3.0089e-2	1.3919	18
peptidyl-prolyl cis-trans isomerase activity	95	2.7280e-23	1.1863e-21	2.0415	240	0.38%	299	7.8420e-3	1.3167	35

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Bejerano Lab, Stanford University

Name	Rank	P-Value	FDR Q-Val	Enrichment	Region Hits	Coverage	Rank	FDR Q-Val	Enrichment	Gene Hits
intramolecular transferase activity	103	6.4868e-22	2.6016e-20	2.5976	135	0.21%	362	3.1525e-2	1.3362	24
NF-kappaB binding	114	1.4476e-20	5.2455e-19	2.0323	212	0.34%	304	8.4708e-3	1.3455	29
translation regulator activity, nucleic acid binding	138	4.3347e-18	1.2976e-16	2.0697	176	0.28%	356	3.0089e-2	1.3919	18
thyroid hormone receptor binding	166	2.6628e-16	6.6264e-15	2.1648	141	0.22%	311	9.1629e-3	1.3919	22
phosphotransferase activity, for other substituted phosphate groups	223	1.6911e-13	3.1326e-12	2.0176	135	0.21%	356	3.0089e-2	1.3919	18

The test set of 63,123 genomic regions picked 15,371 (72%) of all 21,395 genes.
GO Molecular Function has 4,131 terms covering 17,189 (80%) of all 21,395 genes, and 227,341 term - gene associations.
4,131 ontology terms (100%) were tested using an annotation count range of [1, Inf].

Human Phenotype (19 terms)

Table controls:

Export

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Term annotation count: Min:

1

 Max:

Inf

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Visualize this table:

[select one]

Term Name	Binom Rank	Binom Raw P-Value	Binom FDR Q-Val	Binom Fold Enrichment	Binom Observed Region Hits	Binom Region Set Coverage	Hyper Rank	Hyper FDR Q-Val	Hyper Fold Enrichment	Hyper Observed Gene Hits
Abnormality of the mitochondrion	12	1.1672e-48	6.4188e-46	2.0493	518	0.82%	298	3.8977e-8	1.3723	70
Abnormality of mitochondrial metabolism	14	1.3512e-46	6.3689e-44	2.0332	505	0.80%	308	7.1248e-8	1.3717	68
Exercise intolerance	24	1.0246e-38	2.8172e-36	2.0992	385	0.61%	617	5.6753e-4	1.2726	64
Death in infancy	26	5.2193e-37	1.3247e-34	2.1118	362	0.57%	755	4.5822e-3	1.2654	50
Abnormal CSF metabolite level	27	8.7857e-34	2.1473e-31	2.0627	348	0.55%	402	8.4975e-6	1.3463	59
Increased CSF lactate	34	1.9302e-29	3.7462e-27	2.0404	309	0.49%	459	3.3128e-5	1.3422	54
Mitochondrial inheritance	48	6.6953e-25	9.2046e-23	2.0413	258	0.41%	638	8.0186e-4	1.3286	42
Curly hair	67	2.4118e-21	2.3754e-19	2.7200	121	0.19%	985	3.3676e-2	1.3919	16
Abnormal protein N-linked glycosylation	71	2.9241e-20	2.7177e-18	2.4522	138	0.22%	813	7.7965e-3	1.3919	21
Abnormal isoelectric focusing of serum transferrin	81	6.7817e-19	5.5250e-17	2.3940	134	0.21%	852	1.0359e-2	1.3919	20
Abnormal glycosylation	91	4.3004e-18	3.1185e-16	2.2043	152	0.24%	783	5.8137e-3	1.3919	22
Decreased activity of mitochondrial respiratory chain	114	1.1314e-16	6.5490e-15	2.3232	124	0.20%	1,047	4.4111e-2	1.3919	15
Death in childhood	116	1.2732e-16	7.2429e-15	2.2441	133	0.21%	783	5.8137e-3	1.3919	22
Abnormal activity of mitochondrial respiratory chain	121	3.0417e-16	1.6588e-14	2.2491	129	0.20%	985	3.3676e-2	1.3919	16
Abnormal mitochondria in muscle tissue	125	4.4561e-16	2.3525e-14	2.0773	153	0.24%	750	4.3588e-3	1.3919	23
Glioma	159	8.8723e-15	3.6823e-13	2.4049	101	0.16%	1,047	4.4111e-2	1.3919	15
Apathy	198	3.5452e-13	1.1816e-11	2.2683	100	0.16%	908	1.8845e-2	1.3919	18
Diminished motivation	199	3.6175e-13	1.1996e-11	2.2367	103	0.16%	879	1.3981e-2	1.3919	19
Cholelithiasis	238	6.0759e-12	1.6847e-10	2.1369	102	0.16%	852	1.0359e-2	1.3919	20

The test set of 63,123 genomic regions picked 15,371 (72%) of all 21,395 genes.
Human Phenotype has 6,599 terms covering 3,215 (15%) of all 21,395 genes, and 244,972 term - gene associations.
6,599 ontology terms (100%) were tested using an annotation count range of [1, Inf].

Mouse Phenotype Single KO (20+ terms)

Table controls:

Export

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Term annotation count: Min:

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 Max:

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Visualize this table:

[select one]

Term Name	Binom Rank	Binom Raw P-Value	Binom FDR Q-Val	Binom Fold Enrichment	Binom Observed Region Hits	Binom Region Set Coverage	Hyper Rank	Hyper FDR Q-Val	Hyper Fold Enrichment	Hyper Observed Gene Hits
extramedullary hematopoiesis	17	3.1700e-84	1.7099e-81	2.0063	959	1.52%	761	1.2122e-4	1.2401	98
pale liver	37	8.3819e-67	2.0773e-64	2.1872	616	0.98%	784	1.7812e-4	1.2806	69
abnormal circulating tumor necrosis factor level	43	2.0649e-62	4.4035e-60	2.0331	683	1.08%	1,619	3.4859e-2	1.1533	87
increased integument system tumor incidence	51	5.2709e-55	9.4772e-53	2.0861	563	0.89%	576	4.9465e-6	1.3196	73
increased skin tumor incidence	93	4.2957e-38	4.2357e-36	2.1904	343	0.54%	1,101	2.9116e-3	1.2991	42
abnormal cell cycle checkpoint function	102	2.5567e-35	2.2986e-33	2.0575	367	0.58%	630	1.4192e-5	1.3646	50

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Name	Rank	P-Value	FDR Q-Val	Enrichment	Region Hits	Coverage	Rank	FDR Q-Val	Enrichment	Gene Hits
increased hepatocyte apoptosis	112	5.6644e-34	4.6377e-32	2.0284	365	0.58%	807	2.4786e-4	1.3580	40
absent segment of posterior cerebral artery	129	4.9122e-32	3.4919e-30	2.5356	212	0.34%	1,552	2.9349e-2	1.3314	22
increased incidence of tumors by ionizing radiation induction	137	2.5296e-30	1.6931e-28	2.8111	167	0.26%	1,564	2.9472e-2	1.3919	16
increased skin papilloma incidence	149	2.2472e-29	1.3830e-27	2.4163	212	0.34%	1,645	3.7046e-2	1.3286	21
increased hemangioma incidence	162	6.1869e-28	3.5021e-26	2.8456	150	0.24%	1,488	2.2249e-2	1.3919	17
abnormal lipolysis	165	7.6442e-28	4.2483e-26	2.1831	248	0.39%	1,692	4.1653e-2	1.2654	30
increased cellular sensitivity to ultraviolet irradiation	166	1.3099e-27	7.2361e-26	2.4067	200	0.32%	1,103	2.9577e-3	1.3919	24
increased mammary adenocarcinoma incidence	188	5.8822e-25	2.8691e-23	2.4901	168	0.27%	1,645	3.7046e-2	1.3286	21
absent hypoglossal nerve	189	6.2053e-25	3.0107e-23	2.8428	133	0.21%	1,665	3.8545e-2	1.3919	15
abnormal cerebral artery morphology	190	6.3418e-25	3.0608e-23	2.1841	220	0.35%	1,434	1.7665e-2	1.3362	24
increased mammary gland tumor incidence	193	1.0712e-24	5.0896e-23	2.0097	266	0.42%	879	5.7249e-4	1.3553	37
hemolytic anemia	194	1.0764e-24	5.0877e-23	2.1410	228	0.36%	1,611	3.4180e-2	1.2691	31
abnormal oxidative phosphorylation	201	2.0471e-24	9.3394e-23	2.0336	255	0.40%	1,248	7.3994e-3	1.2901	38
abnormal circulating interleukin-1 beta level	203	2.6206e-24	1.1838e-22	2.0022	264	0.42%	1,627	3.6823e-2	1.2491	35

The test set of 63,123 genomic regions picked 15,371 (72%) of all 21,395 genes.
Mouse Phenotype Single KO has 9,170 terms covering 9,466 (44%) of all 21,395 genes, and 551,620 term - gene associations.
9,170 ontology terms (100%) were tested using an annotation count range of [1, Inf].

Mouse Phenotype (20+ terms)

Table controls:

Export

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20

Set

 Term annotation count: Min:

1

 Max:

Inf

Set

 Visualize this table:

[select one]

Term Name	Binom Rank	Binom Raw P-Value	Binom FDR Q-Val	Binom Fold Enrichment	Binom Observed Region Hits	Binom Region Set Coverage	Hyper Rank	Hyper FDR Q-Val	Hyper Fold Enrichment	Hyper Observed Gene Hi
increased hemangioma incidence	127	1.8033e-39	1.3595e-37	2.3472	307	0.49%	1,153	2.9086e-4	1.3919	31
increased cellular sensitivity to ultraviolet irradiation	133	5.6526e-39	4.0694e-37	2.5715	254	0.40%	1,189	3.9281e-4	1.3919	30
increased incidence of tumors by ionizing radiation induction	166	4.7017e-33	2.7120e-31	2.6988	196	0.31%	1,727	1.0325e-2	1.3919	19
pancytopenia	171	3.1952e-32	1.7891e-30	2.2238	278	0.44%	1,482	2.9809e-3	1.3470	30
absent segment of posterior cerebral artery	174	4.9122e-32	2.7031e-30	2.5356	212	0.34%	1,978	2.4045e-2	1.3314	22
absent hypoglossal nerve	243	6.2053e-25	2.4451e-23	2.8428	133	0.21%	2,091	3.2048e-2	1.3919	15
abnormal cerebral artery morphology	244	6.3418e-25	2.4887e-23	2.1841	220	0.35%	1,824	1.4502e-2	1.3362	24
hemolysis	258	4.9149e-24	1.8240e-22	2.0990	232	0.37%	1,562	4.6019e-3	1.3146	34
hemolytic anemia	265	1.3555e-23	4.8977e-22	2.0743	234	0.37%	1,963	2.2832e-2	1.2726	32
decreased mean platelet volume	276	5.8103e-23	2.0157e-21	2.5680	145	0.23%	1,727	1.0325e-2	1.3919	19
increased cellular sensitivity to alkylating agents	280	1.2715e-22	4.3480e-21	2.1703	201	0.32%	1,675	7.6455e-3	1.3919	20
decreased cellular sensitivity to ultraviolet irradiation	292	7.5024e-22	2.4601e-20	2.3576	162	0.26%	2,091	3.2048e-2	1.3919	15
abnormal DNA replication	302	1.6155e-21	5.1221e-20	2.1155	202	0.32%	1,751	1.1242e-2	1.3384	25
shiny skin	306	3.4798e-21	1.0889e-19	2.0213	222	0.35%	2,116	3.4778e-2	1.2654	30
increased cellular sensitivity to DNA damaging agents	325	1.8167e-20	5.3522e-19	2.0358	210	0.33%	1,547	4.2696e-3	1.3919	22
abnormal small intestinal crypt cell proliferation	331	2.4891e-20	7.2002e-19	2.0788	198	0.31%	1,547	4.2696e-3	1.3919	22
embryonic lethality before implantation, incomplete penetrance	376	8.1691e-19	2.0803e-17	2.1279	172	0.27%	2,067	3.0784e-2	1.3286	21
abnormal Kupffer cell morphology	394	2.5519e-18	6.2016e-17	2.2314	150	0.24%	1,727	1.0325e-2	1.3919	19
increased kidney tumor incidence	436	2.9840e-17	6.5531e-16	2.1295	156	0.25%	1,986	2.4235e-2	1.3919	16
increased incidence of tumors by UV-induction	472	2.3461e-16	4.7594e-15	2.8194	85	0.13%	2,183	4.2738e-2	1.3919	14

The test set of 63,123 genomic regions picked 15,371 (72%) of all 21,395 genes.
Mouse Phenotype has 9,575 terms covering 9,654 (45%) of all 21,395 genes, and 705,265 term - gene associations.
9,575 ontology terms (100%) were tested using an annotation count range of [1, Inf].

https://great.stanford.edu/great/public/cgi-bin/greatWeb.php

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